

SCALE FOR PROJECT PYTHON MODULE (/PROJECTS/PYTHON-MODULE-10)

You should evaluate 1 student in this team



Git repository

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Introduction

- Remain polite, courteous, respectful and constructive throughout the evaluation process. The well-being of the community depends on it.
- Work with the person (or the group) being evaluated to identify any potential dysfunctions in the work. Take the time to discuss and debate the problems you have identified.
- You must consider that there may be differences in how your peers have understood the project's instructions and the scope of its functionalities. Always keep an open mind and grade them as honestly as possible. The pedagogy is valid only if peer evaluation is conducted seriously.

Guidelines

- Only grade the work that is in the student's or group's Git repository.
- Double-check that the Git repository belongs to the student or the group. Ensure that the work is for the relevant project and also check that "git clone" is used in an empty folder.
- Check carefully that no malicious aliases were used to fool you and make you evaluate something other than the content of the official repository.
- To avoid any surprises, carefully check that both the evaluating and the evaluated students have reviewed the possible scripts used to facilitate the grading.
- If the evaluating student has not completed that particular project yet, it is mandatory for this student to read the entire subject prior to starting the defence.
- Use the flags available on this scale to signal an empty repository, a non-functioning program, a norm error, cheating, etc. In these cases, the grading is over and the final grade is 0 (or -42 in case of cheating). However, with the exception of cheating, you are encouraged to continue to discuss your work (even if you have not finished it) in order to identify any issues that may have caused this failure and avoid repeating the same mistake in the future.
- Remember that during the defence, no unexpected, premature, uncontrolled termination of the program occurs; otherwise, the final grade is 0. Use the appropriate flag.
Unless stated in the scale, you should never need to edit any file except the configuration file, if it exists.
If you want to edit a file, take the time to explain the reasons to the evaluated student and make sure both of you agree to this.

Attachments

 subject.pdf (<https://cdn.intra.42.fr/pdf/pdf/203217/en.subject.pdf>)

 generator.tar.gz (<https://cdn.intra.42.fr/document/document/48270/generator.tar.gz>)

Preliminaries

Basics

- The review is done in the presence of the reviewed learner. This is how everyone progresses: by interacting with others.
 - No submitted work: 0, the review is over.
 - As soon as an exercise is non-functional, the review stops. You can look at the code o exercises, but they will not be graded.
- Check that the following files are present:
 - ex0/lambda_spells.py
 - ex1/higher_magic.py
 - ex2/scope_mastersies.py
 - ex3/functools_artifacts.py
 - ex4/decorator_mastery.py If any required files are missing, the peer-review stops here
- Verify that the code runs without errors and demonstrates the required concepts.
- Test both valid and invalid scenarios to ensure proper validation.
- Look for proper use of fonctionnal features: fonctionnal patterns, anonymous functions, lambd functools...
- Ensure all dependencies are installed via proper package management in a virtual environment
- Ensure the learner avoided Global variables and File I/O operations.

 Yes

 No

Exercises

Exercise 0 - Lambda Sanctum

Check that lambda_spells.py is properly implemented:

- Does artifact_sorter() use sorted() with a lambda to sort by 'power' descending?
- Does power_filter() use filter() with a lambda to filter mages by minimum power?
- Does spell_transformer() use map() with a lambda to add * prefix and suffix?
- Does mage_stats() use lambdas with max(), min() for statistics?

Test the functions with sample data:

Using data_generator.py in attachments to generate sample data:

- Create test artifacts and verify sorting works correctly
- Test power filtering with different minimum values
- Verify spell transformation adds the correct decorations to * spells *
- Check that statistics calculation returns correct max, min, and average

Verify that lambda expressions are used appropriately (not def functions for simple operations). If everything works correctly, tick "Yes." Otherwise, tick "No" and the peer-review is over.

 Yes

 No

Exercise 1 - Higher Realm

Check that higher_magic.py demonstrates higher-order functions:

- Does spell_combiner() return a function that calls both input functions?
- Does power_amplifier() return a function that multiplies the base spell result?
- Does conditional_caster() return a function that checks condition before casting?
- Does spell_sequence() return a function that executes all spells in order?

Test the higher-order function behavior:

Using data_generator.py to have sample data for spells, targets and power values :

- Create simple test functions and verify combination works
- Test power amplification with different multipliers

- Verify conditional casting works when both true and false conditions are met
- Check that spell sequences execute all spells and return results list

Verify that functions are treated as first-class citizens (passed as arguments, returned from function). If everything works correctly, tick "Yes." Otherwise, tick "No" and the peer-review is over.

☒ Yes

☐ No

Exercise 2 - Memory Depths

Check that `scope_mysteries.py` demonstrates lexical scoping and closures:

- Does `mage_counter()` return a function that maintains count state?
- Does `spell_accumulator()` return a function that accumulates power over calls?
- Does `enchantment_factory()` return functions that remember their enchantment type?
- Does `memory_vault()` return a dict with store/recall functions sharing private storage?

Test closure behavior:

If you need to, generate sample data using `generato_data.py`

- Create multiple counters and verify they maintain separate state
- Test power accumulation with different initial values Example: create with `intial_power=0`. Ca 20, then 30.
- Create different enchantment factories and verify they work independently Example: create "Frozen", and apply to same item "Axe". Expect different results.
- Test memory vault storage and recall functionality. Example: Store 3 values, recall all 3, recall reviewee to explain what happens.

Verify that closures properly capture and maintain their lexical environment.

Verify there is no use of 'global' keyword, and no class definitions.

If everything works correctly, tick "Yes." Otherwise, tick "No" and the peer-review is over.

☒ Yes

☐ No

Exercise 3 - Ancient Library

Check that `functools_artifacts.py` properly uses the `functools` module:

Spell Reducer

- Does `spell_reducer()` use `functools.reduce` with operator module functions?
- Does `spell_reducer()` use `operator.add` and `operator.mul` for "add" and "multiply"?
- Ask the learner to explain how are handled "min" and "max". Do you know other solutions?
- Test reduce with different operations (add, multiply, max, min)

Partial Enchanter

- Does `partial_enchanter()` use `functools.partial` to create specialized functions (not lambdas w
- Verify partial application creates working specialized functions
- Test with any function(`power`, `element`, `target`) the result, with different powers
- Verify `type(result['fire_enchant'])` is `functools.partial`

Memoized Fibonacci

- Does `memoized_fibonacci()` use `functools.lru_cache` for memoization?
- Test fibonacci memoization improves performance on repeated calls. If you don't know how, to show you how.
- Verify: `memoized_fibonacci.cache_info()` returns a `CacheInfo` with `hits > 0` after calling `fib(10)`

Spell Dispatcher

- Does `spell_dispatcher()` use `functools singledispatch` for type-based dispatch?
- Check single dispatch handles different types appropriately. Try also with a float.

Ask the learner

- "What happens if you call reduce on an empty list without an initializer?"
- "What is the difference between memoization and simply storing results in a global dict?"

If ALL tests pass and `functools` features are used (not manual reimplementation), tick "Yes." Otherwise, the peer-review is over.

☒ Yes

☐ No

Exercise 4 - Master's Tower

As usual, you can generate sample data for spells, names, Mages using `data_generator.py`

Check that `decorator_mastery.py` demonstrates decorator mastery:

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- Does `spell_timer()` create a decorator that measures execution time?
- Does `power_validator()` create a parameterized decorator for validation?
- Does `retry_spell()` create a decorator that retries failed functions?
- Does `MageGuild` class properly use `@staticmethod`?
- Does `cast_spell` uses `power_validator` with `min_power=10` ?

Test decorator functionality:

- Test `spell_timer` with a function and verify timing output
- Test `power_validator` with valid and invalid power levels
- Test `retry_spell` with functions that fail and eventually succeed
- Verify `staticmethod` works without instance
- Test `MageGuild` with valid and invalid names
- Test what `cast_spells` returns, on success and on failure. Try different powers.

Check that `functools.wraps` is used to preserve function metadata.

Verify that decorators properly transform function behavior.

If everything works correctly, tick "Yes." Otherwise, tick "No" and the peer-review is over.

☒ Yes

☐ No

Code Quality and Understanding

Code Quality and Understanding

Evaluate the learner's understanding of functional programming concepts:

- Can the learner explain the difference between `lambda` and `def` functions?
- Can the learner explain what makes functions "first-class citizens"?
- Does the learner understand how closures capture lexical scope?
- Can the learner explain the benefits of `functools` utilities?
- Does the learner understand how decorators transform functions?

Check for code quality:

- Is the code written in Python 3.10 or higher?
- Does the code adhere to the `flake8` linter standards (no errors)?
- Are type hints `REQUIRED` for all functions and methods (parameters and return values)?
- Docstrings are `NOT` required for this module.
- Is the code clean and readable?
- Are functional programming patterns used correctly?
- Does the code demonstrate understanding rather than just copying?

Discuss the concepts with the learner to assess their understanding.

If the learner demonstrates good understanding of functional programming, tick "Yes."

☒ Yes

☐ No

Ratings

Don't forget to check the flag corresponding to the defense

☒ Ok

☐ Outstanding project

☐ Empty work

☐ Incomplete work

☐ Norme

☐ Cheat

☐ Crash

☐ Conce

☐ Leaks

☐ Forbidden function

☐ Can't support / explain

Conclusion

Leave a comment on this evaluation (2048 chars max)

Finish evaluation

